Project R-DAS: Onboard Deep-Space Autonomy Suite for ESA Hera

ESA Open Space Innovation Platform (OSIP) | Autonomous Software Experiments on Hera

EXECUTIVE SUMMARY & MISSION IMPACT:
radixal s.r.o. (Brno, Czech Republic) submits a comprehensive 6-proposal suite, the Radixal Deep-Space Autonomy Suite (R-DAS), under the ESA OSIP Call for Ideas. Engineered for bare-metal execution on the Frontgrade Gaisler GR712RC (LEON3 @ 50 MHz) Core 1 sandbox of ESA's Hera spacecraft during the August 2027 Extended Mission.

• Key Technological Innovations: Autonomous detection and cataloging of asteroid impact craters and surface depressions via INT8 Micro-CNN & PALT laser fusion, lossless CDF 5/3 integer wavelet compression (-82.2% bandwidth), in-situ 3D shape mesh building, and passive gravity parameter (GM) inversion.

• In-Flight Benchmark & Ramses Synergy: Executes an in-flight Shadow-Mode GNC benchmark comparing onboard maneuver recommendations against ESOC Flight Dynamics ground truth, establishing TRL 8 flight qualification for the 2029 ESA Ramses mission to asteroid (99942) Apophis.

The 6 Proposals of the R-DAS Suite (Covering All 6 ESA Contest Categories)

Module / Identifier	ESA Category	Primary Function & In-Flight Objective on Hera	Engineering Budget
1. ARGOS-AI (Flagship)	Cat. 4: Edge AI	Impact crater detection in INT8 Micro-CNN + PALT laser altimeter metric sizing in meters	2.39 s WCET | 142 kB Scratchpad (+1.04 MB Camera Frame)
2. DEEP-WAVE	Cat. 2: Compression	Reversible CDF 5/3 integer wavelet compression achieving -82.2% downlink bandwidth reduction	2.39 s WCET | 38 kB Scratchpad (Tile-based streaming)
3. AURA-GNC	Cat. 1: Autonomy & GNC	Shadow-Mode 3D landmark mesh building from craters, GM gravity inversion & ESOC validation	3.80 s WCET | 96 kB Scratchpad (9-state EKF filter)
4. AEGIS-FDIR	Cat. 5: Resilience & FDIR	Operationalizing ESTEC HERA-IoD research: Quantized Isolation Forest telemetry anomaly detector	0.12 s WCET | 18 kB Scratchpad (Quantized trees)
5. ARES-Planner	Cat. 3: Operations	Autonomous integer CSP solver for multi-payload observation scheduling (+35% science return)	1.40 s WCET | 43 kB Scratchpad (Integer solver)
6. CHRONOS	Cat. 6: Science / Astro	Onboard aperture photometry & Dimorphos lightcurve tracking (Ondrejov Observatory synergy)	0.85 s WCET | 29 kB Scratchpad (Aperture phot.)

Technical Feasibility & Spacecraft Safety (Zero Flight Risk)

Engineered strictly in deterministic ANSI C99 without dynamic memory allocation (0 malloc, < 24 kB stack). Stated memory budgets represent static algorithm scratchpads; camera frame buffers (1.04 MB / image) are processed via streaming DMA. The codebase was empirically validated on 2,400+ real Asteroid Framing Camera (AFC) calibration images inside the QEMU LEON3 SPARC V8 emulator. Execution is confined to the isolated Core 1 bare-metal sandbox without actuator authority, ensuring absolute zero risk to Core 0 flight software.

Proposing Entity: radixal s.r.o. & Leadership Triad

radixal s.r.o. (est. 2016 in Brno) holds a 10-year track record in safety-critical systems (AK Signal / SIL railway), satellite image wavelet pipelines for ESA Copernicus Sentinel-2 (Spacemetric AB / Sweden & Norway), defense systems (URC Systems), and national infrastructure.

• Bc. Viktor Lostak (Principal Investigator & Lead Architect): Lead architect and designer of mathematical algorithms.
• Ing. Petr Slepicka (Engineering Lead & Delivery Director): Lead flight software engineer, MISRA-C static verification, and ECSS QA.
• Mgr. David Riedl (Executive Director & Governance): Project governance, legal, and ESA institutional compliance.

PRINCIPAL INVESTIGATOR & CONTACT:
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